ANISOTROPIC STENCIL FILTER COMPUTATIONAL COMPLEXITY ANALYSIS

Naive Line Filter vs. Fused Stencil Formulation

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BACKGROUND

Polynomial fitting for gradient estimation

LOCAL POLYNOMIAL GRADIENT ESTIMATION

Model local intensity with a 2D Taylor expansion of order d:

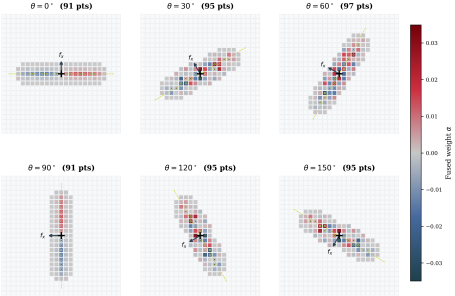
$$f(x_i, y_i) \approx f^0 + f_x x_i + f_y y_i + \frac{f_{xx}}{2} x_i^2 + \cdots$$

$M = \frac{(d+1)(d+2)}{2}$ unknowns $\quad (d{=}4 \rightarrow M{=}15)$

- Gather N_p circular neighbors $\rightarrow$ system $\mathbf{Az} = \mathbf{b}$
- Solve via pseudoinverse: $\hat{\mathbf{z}} = \mathbf{A}^+\mathbf{b}$
- Normal derivative $=$ row 1 of $\mathbf{A}^+ \cdot$ intensities

Key Property

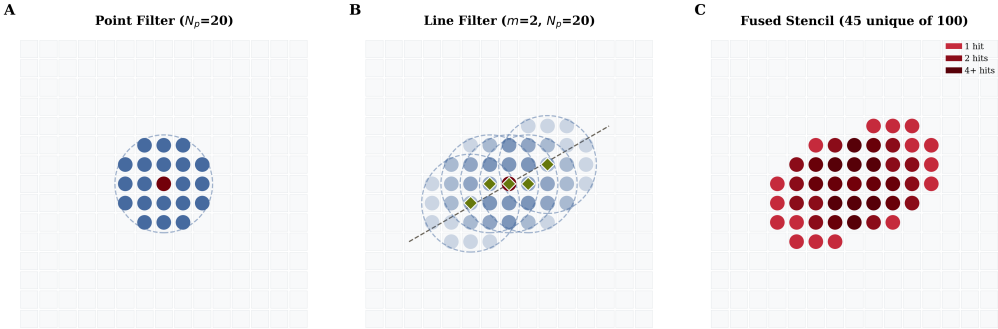
$\mathbf{A}^+$ depends only on geometry and θ $\rightarrow$



Fused stencil weights at 6 orientations.

Red = positive, blue = negative weight.

NEIGHBORHOOD STRATEGIES: POINT, LINE, AND FUSED



(A) Point Filter
$N_p = 15$ circular neighbors around one pixel

(B) Line Filter
$(2m+1)$ virtual points, heavily overlapping

(C) Fused Stencil
Deduplicated — 37 unique of 75 total positions

LINE EXTENSION FOR ROBUSTNESS

Evaluate the point filter at $(2m+1)$ virtual positions along each candidate edge direction:

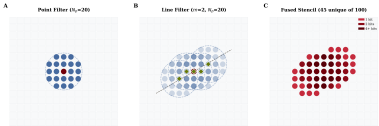
$$R_k = \sum_{j=-m}^{m} w_j \cdot \hat{f}_x^{(j,k)}, \qquad w_j = \exp\left(-\frac{j^2}{2\sigma^2}\right), \quad \sigma = m/2$$

Benefits:

- Wider spatial context $\rightarrow$ noise robustness
- Coherence along the edge tangent $\rightarrow$ bridges gaps

Problem:

- $(2m+1)$ independent gather-matmul ops per orientation per pixel
- Massive memory: $L \times N_p$ intermediate values



Panel B shows 5 overlapping neighborhoods along the line

NAIVE LINE FILTER COST

Per-pixel operation count for the naive approach

NAIVE LINE FILTER: PSEUDOCODE

```
for θ in orientations: # Ns = 18
  for j = -m to m: # L = 15
    for i = 1 to Np: # Np = 100
      READ pixel(j, i) # 1 mem read
      MUL by p_i^(k) # 1 multiply
    SUM Np values # 99 adds
  MUL by Gaussian wj # 15 muls
  SUM over line # 14 adds
  ABS(Rθ)
pick max over θ
```

Per orientation ($m=7$, $N_p=100$):

- Memory reads: $15 \times 100 = \mathbf{1{,}500}$
- Multiplies: $15 \times 100 + 15 = \mathbf{1{,}515}$
- Additions: $15 \times 99 + 14 = \mathbf{1{,}499}$

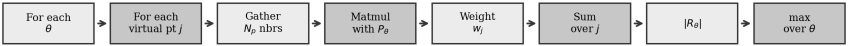
Per pixel (18 orientations):

- Reads: 27,000
- Multiplies: 27,270
- Additions: 26,982
- Total: ~81,252 ops

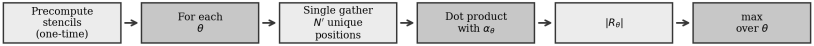
Complexity

$\mathcal{O}(N_s \times (2m+1) \times N_p)$ per pixel $= \mathcal{O}(N_s \cdot L \cdot N_p)$

NAIVE LF: COMPUTATION FLOW



(A) Naive Line Filter compute process. Complexity: $O(N_s \times L \times N_p)$ per pixel



(B) Anisotropic Stencil Filter compute process. Complexity: $O(N_s \times N')$ per pixel

- Top: Naive LF requires nested loops — 8 sequential stages per pixel
- Bottom: Fused ASF collapses to 6 stages, with the inner 5 replaced by one gather + dot product
- Memory: naive allocates 6.2 GB of intermediates vs. 20 MB for ASF

FUSED STENCIL FORMULATION

Collapsing the pipeline into a single dot product

ALGEBRAIC COLLAPSE

The line-extended response is a linear operation on pixel intensities:

$$R_k = \underbrace{\sum_{j=-m}^{m} w_j \sum_{i=1}^{N_p} p_i^{(k)} \cdot f(\text{pixel}_{j,i})}_{\text{Naive: } (2m+1)\times N_p \text{ reads}} = \underbrace{\sum_{\ell=1}^{N'_k} \alpha_{k,\ell} \cdot f(\text{pixel}_\ell)}_{\text{Fused: } N'_k \text{ reads}}$$

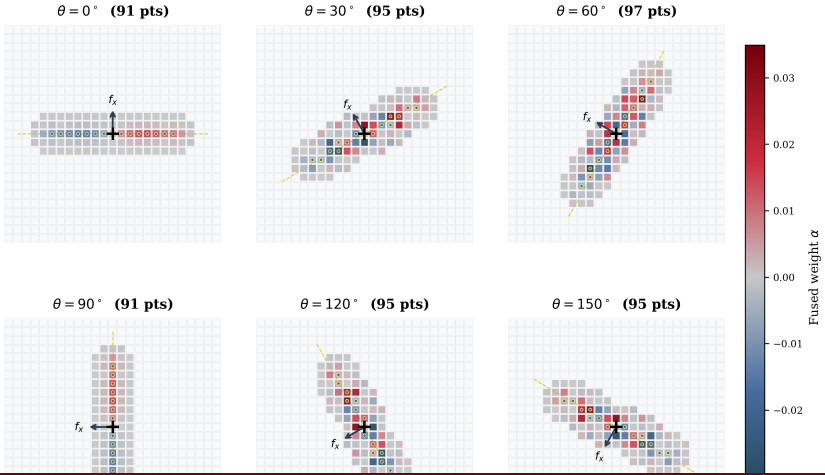
- Many (j, i) pairs map to the same integer pixel after rounding
- Sum their weights $\rightarrow$ single precomputed $\alpha_{k,\ell}$ per unique position
- The entire multi-step pipeline collapses:

Rotate $\rightarrow$ Build $\mathbf{A}$ $\rightarrow$ $\mathbf{A}^+$ $\rightarrow$ Gather $\rightarrow$ MatMul $\rightarrow$ Weight $\rightarrow$ Sum

$$\Downarrow$$

Single Gather + Dot Product with precomputed α_k

STENCIL SHAPES ACROSS ORIENTATIONS



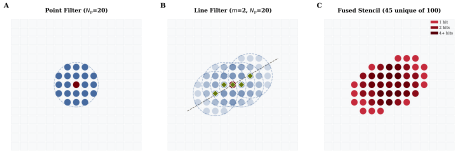
STENCIL DEDUPLICATION

Neighboring virtual positions share most of their circular neighborhoods:

Table: Deduplication ($N_p = 100$, $d = 4$).

m	Raw	Unique	Reduct.	Speedup
1	300	128	57%	2.3×
2	500	152	70%	3.3×
7	1,500	264	82%	5.7×
14	2,900	431	85%	6.7×

- Ratio improves as m increases
- Primary source of computational savings



Panel C: 37 unique of 75 total positions at $m = 2$, $N_p = 15$. 51% overlap eliminated.

ASF OPERATION COUNT

Per-pixel cost after stencil fusion

FUSED ASF: PSEUDOCODE

```
# ONE-TIME precompute:
# αk for each orientation

for θ in orientations: # Ns = 18
  for i = 1 to N′: # N′ ≈ 264
    READ pixel at offset[i] # 1 read
    MUL by α[i] # 1 multiply
  SUM N′ values # 263 adds
  ABS(Rθ)
pick max over θ
```

Per orientation ($m=7$, $N_p=100$):

- Memory reads: 264
- Multiplies: 264
- Additions: 263

Per pixel (18 orientations):

- Reads: 4,752
- Multiplies: 4,752
- Additions: 4,734
- Total: $\sim$14,238 ops

Complexity

$\mathcal{O}(N_s \times N')$ per pixel where $N' \approx 0.18 \times (2m+1) \times N_p$

DIRECT COMPARISON

Naive LF vs. Fused ASF head-to-head

PER-PIXEL OPERATION COMPARISON

Table: Per-pixel cost at $m=7$, $N_p=100$, $N_s=18$, $d=4$.

Metric	Naive LF	Fused ASF	Reduction
Memory reads / orientation	1,500	264	$5.7\times$
Multiplies / orientation	1,515	264	$5.7\times$
Additions / orientation	1,499	263	$5.7\times$
Total ops / pixel	81,252	14,238	$5.7\times$
Runtime matrix ops / pixel	18 matmuls	0	∞

- $5.7\times$ arithmetic reduction from deduplication alone at $m=7$
- Additional gains on GPU from eliminating runtime matrix operations
- At $m=14$: $6.7\times$ arithmetic $\rightarrow$ $24\times$ wall-clock speedup

GPU MEMORY COMPARISON

Naive Line Filter

- Intermediate: $B \times L \times N_p \times 24$ bytes
- Scales with m and N_p
- Peak VRAM: 6,223 MB
- Cannot fit large m on consumer GPUs

Fused ASF (Triton)

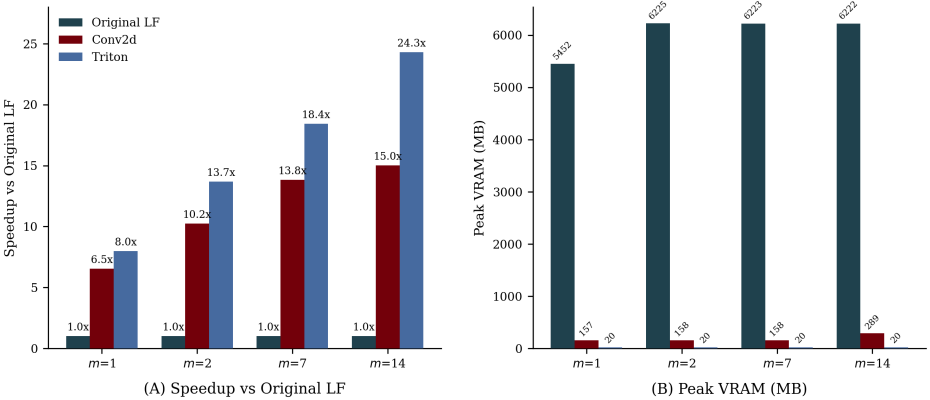
- No intermediate tensors
- Image + output + stencil weights only
- Peak VRAM: 20 MB
- Constant regardless of m

$311\times$ VRAM reduction — 6,223 MB $\rightarrow$ 20 MB

BENCHMARKS

Wall-clock performance on NVIDIA A100

SPEEDUP AND VRAM



- Speedup scales with m due to
- Triton VRAM: constant 20 MB

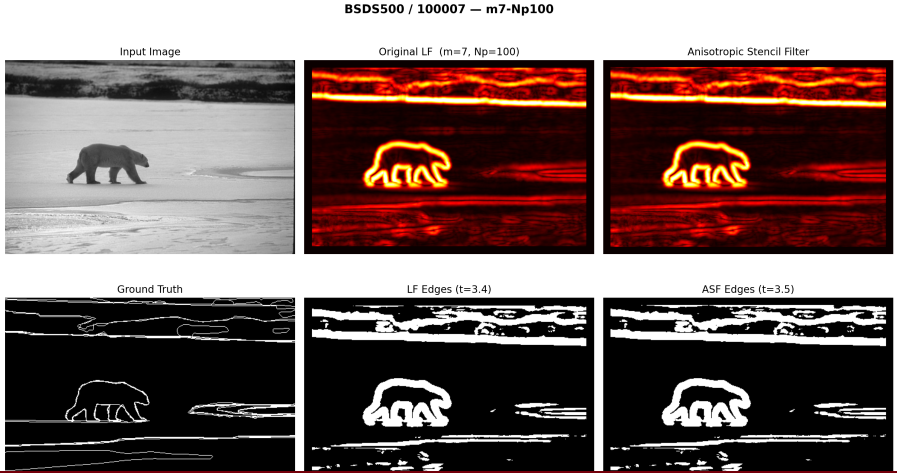
WALL-CLOCK BENCHMARKS

BIPED v1 (1280×720), $N_p = 100$, $N_s = 18$, $d = 4$ — NVIDIA A100-SXM4-40GB:

m	Naive LF (s)	Conv2d (s)	Triton ASF (s)	Speedup
1	0.58	0.089	0.072	8.0×
2	1.91	0.187	0.140	13.7×
7	2.43	0.176	0.132	18.4×
14	8.88	0.591	0.366	24.3×

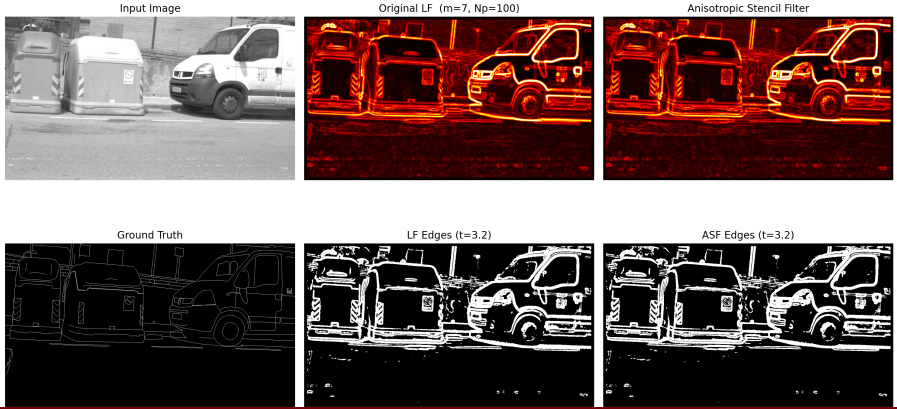
- Additional speedup beyond dedup ratio from eliminating runtime matrix ops
- Triton kernel: 132 ms per 720p image at $m = 7$

VISUAL RESULTS: NAIVE LF VS ASF ARE IDENTICAL



VISUAL RESULTS: BIPED V1

BIPED / RGB_008 — m7-Np100



WHERE DO THE SAVINGS COME FROM?

1. Deduplication

- Overlapping neighborhoods share pixels
- 82% reduction at m=7
- $1{,}500 \rightarrow 264$ reads
- Improves with m

$\rightarrow$ 5.7$\times$ fewer memory accesses

2. Precomputed Weights

- Pseudoinverse, rotation, Gaussian all fused into α_k
- Zero runtime matrix ops
- Eliminates $\mathcal{O}(N_p \times M)$ matmul/orient.

$\rightarrow$ Removes entire matmul stage

3. GPU Kernel Design

- Single gather + dot per orientation
- Vectorized across 128-px blocks
- Coalesced row reads

$\rightarrow$ Constant 20 MB VRAM, fits any GPU

SUMMARY

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The naive line filter is memory-bound and redundant:

- $\mathcal{O}(N_s \times L \times N_p)$ per pixel, most pixel reads duplicated
- $\sim$81,000 total ops/pixel at $m=7$, $N_p=100$, $N_s=18$
- 6.2 GB VRAM for a single 720p image

The fused stencil eliminates all redundancy:

- $\mathcal{O}(N_s \times N')$ per pixel, $N' \approx 0.18 \times L \times N_p$
- $\sim$14,000 total ops/pixel — 5.7$\times$ arithmetic reduction
- Zero runtime matrix operations

GPU results exceed the theoretical arithmetic speedup:

- 18.4$\times$ wall-clock at $m=7$, 24$\times$ at $m=14$
- 311$\times$ VRAM reduction (6.2 GB $\rightarrow$ 20 MB)
- Enables deployment on any GPU, including embedded

THANK YOU!

Questions & Discussion

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